

Lecture 15. Cross-Validation

COMP90051 Statistical Machine Learning

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This lecture

- Motivation
- Performance metrics
- Cross-validation
- Importance of variability
- Examples and misunderstandings

Motivation

Our goal in machine learning?

This lecture

- Our goal in machine learning?
 - * Have a perfect model?
 - Not necessarily clear what that means
 - Even simple problems cannot be perfectly solved (see slide 10)
 - * Use algorithms that will perform well in unseen data
 - How to measure performance?
 - How to use unseen data?
- Related aspects:
 - * Compare different algorithms in terms of performance from limited available data
 - * Choose hyperparameters for an algorithm, e.g., C for soft-margin support vector machines

Performance metrics

How to measure performance?

Performance metrics

- Given a dataset $D = \{x_1, y_1, \dots, x_n, y_n\}$ of n samples
- Assume that for a data point x_i we predict $g(x_i)$
- Some metrics in regression:
 - * Mean squared error: $MSE(g) = \frac{1}{n} \sum_{i=1}^n (g(x_i) - y_i)^2$
 - * Root mean squared error: $RMSE(g) = \sqrt{MSE(g)}$
 - * Mean absolute error: $\frac{1}{n} \sum_{i=1}^n |g(x_i) - y_i|$

Performance metrics

- True Positive (TP), True Negative (TN), False Positive (FP), False Negative (FN)

		True Label	
		+1	-1
Predicted Label	+1	TP	FP
	-1	FN	TN

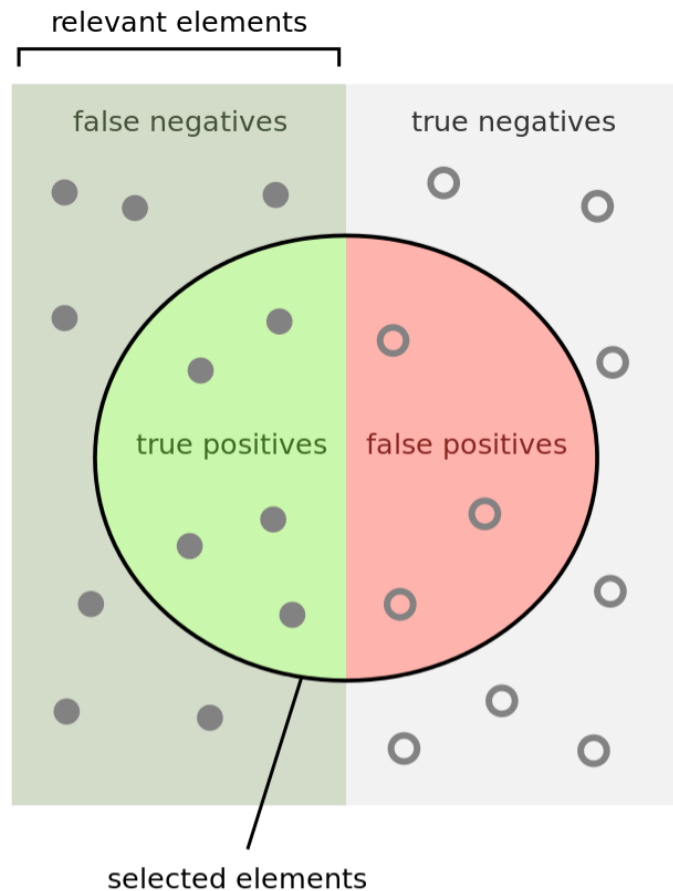
- Some metrics classification:
 - * Accuracy $(TP + TN) / (TP + FP + FN + TN)$
 - * Error $(FP + FN) / (TP + FP + FN + TN)$
 - * Recall / Sensitivity $TP / (TP + FN)$
 - * Precision $TP / (TP + FP)$
 - * Specificity $TN / (TN + FP)$
 - * F1-score $2 \text{ Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

Performance metrics

- Precision and recall are typically in an inverse relationship:
 - * The classifier has high Precision, but low Recall
 - * The classifier has high Recall, but low Precision
- Similar for sensitivity and specificity
- Use jointly:
 - * (Precision, Recall)
 - * (Sensitivity, Specificity)

Precision and recall

- Idea comes from information retrieval



How many selected items are relevant?

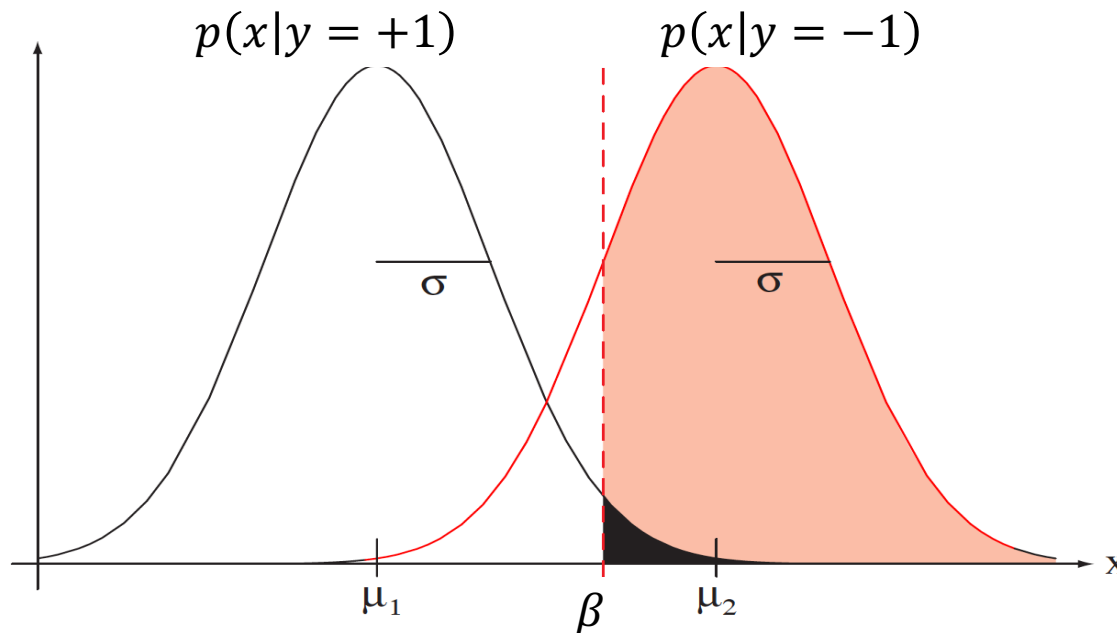
$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

Sensitivity and specificity

- Idea comes from signal detection theory
 - * Assume Gaussian distributions $p(x|y = +1) = N(\mu_1, \sigma^2)$
 $p(x|y = -1) = N(\mu_2, \sigma^2)$



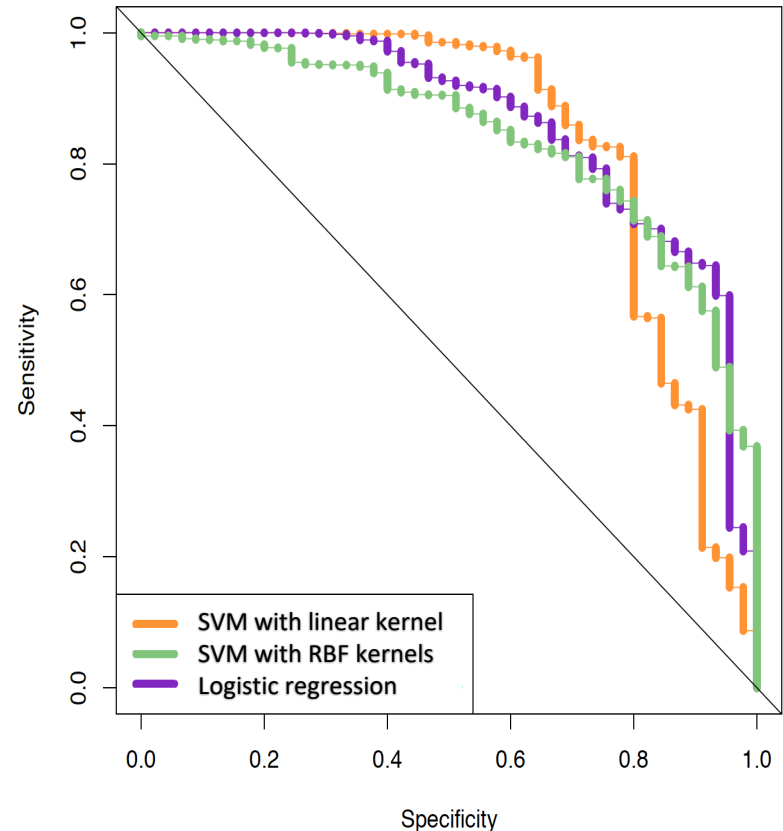
Classifier:

$$g(x) = \begin{cases} +1, & \text{if } x < \beta \\ -1, & \text{if } x \geq \beta \end{cases}$$

- * By sliding the offset β we get different (TP, FP, TN, FN) and thus, different sensitivity and specificity

Receiver Operating Characteristic (ROC)

- By varying the offset (or a hyperparameter) for a classifier (e.g., SVMs, logistic regression) we can get different:
 - * Sensitivity
 - * Specificity
- Summarized with an Area Under the Curve (AUC)
 - * Random: 0.5
 - * Perfect classifier: 1



Chance level error

- Consider a classification problem:
 - * 2 classes, 50% of samples each class
- Consider three classifiers:
 - * A classifier that always predicts class +1 gets 50% error
 - * A classifier that always predicting class -1 gets 50% error
 - * A random classifier that would flip a coin on every prediction gets 50% error (approximately)
- Can a classifier get more than 50% error?
 - * Yes, for test error (e.g., we could get less than 50/100 marks in an exam with true/false questions)

Other loss functions

- Consider classification with health data
 - Let +1 be “diseased patient” and -1 be “healthy patient”
 - Predicting a healthy patient as diseased (e.g., unnecessary medical procedures) has different consequences than predicting a diseased patient as healthy (e.g., missing early treatment)

		True Label	
		+1	-1
Predicted Label	+1	0	1
	-1	1	0



		True Label	
		+1	-1
Predicted Label	+1	0	1
	-1	10	0

0/1 loss:

$$\frac{1}{n} \sum_{i=1}^n 1(g(x_i) \neq y_i)$$

$$\frac{1}{n} \sum_{i=1}^n \text{Cost}(g(x_i), y_i)$$

Mini Summary

- Different performance metrics
 - * Choose the most suitable ones for your problem
- More things to consider:
 - * learning curves (e.g., number of samples versus accuracy)
 - * confusion matrices
 - * micro-averaging or macro-averaging for multi-class classification
 - * beyond supervised machine learning, one can use performance metrics based on the log-likelihood (e.g., probabilistic graphical models)

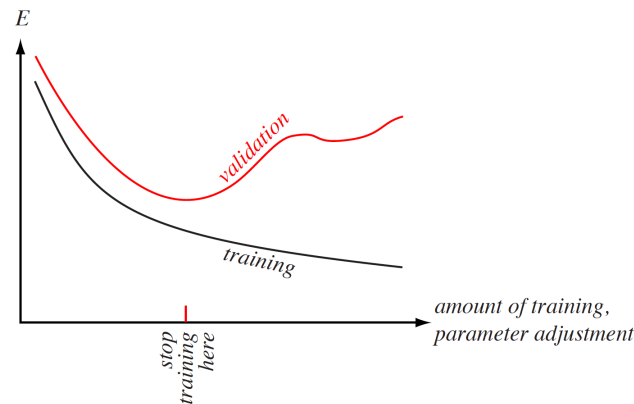
Next: Using unseen data

Cross-validation (or using “unseen” data)

Wait for many years to receive new
samples or simulate

Using “unseen” data

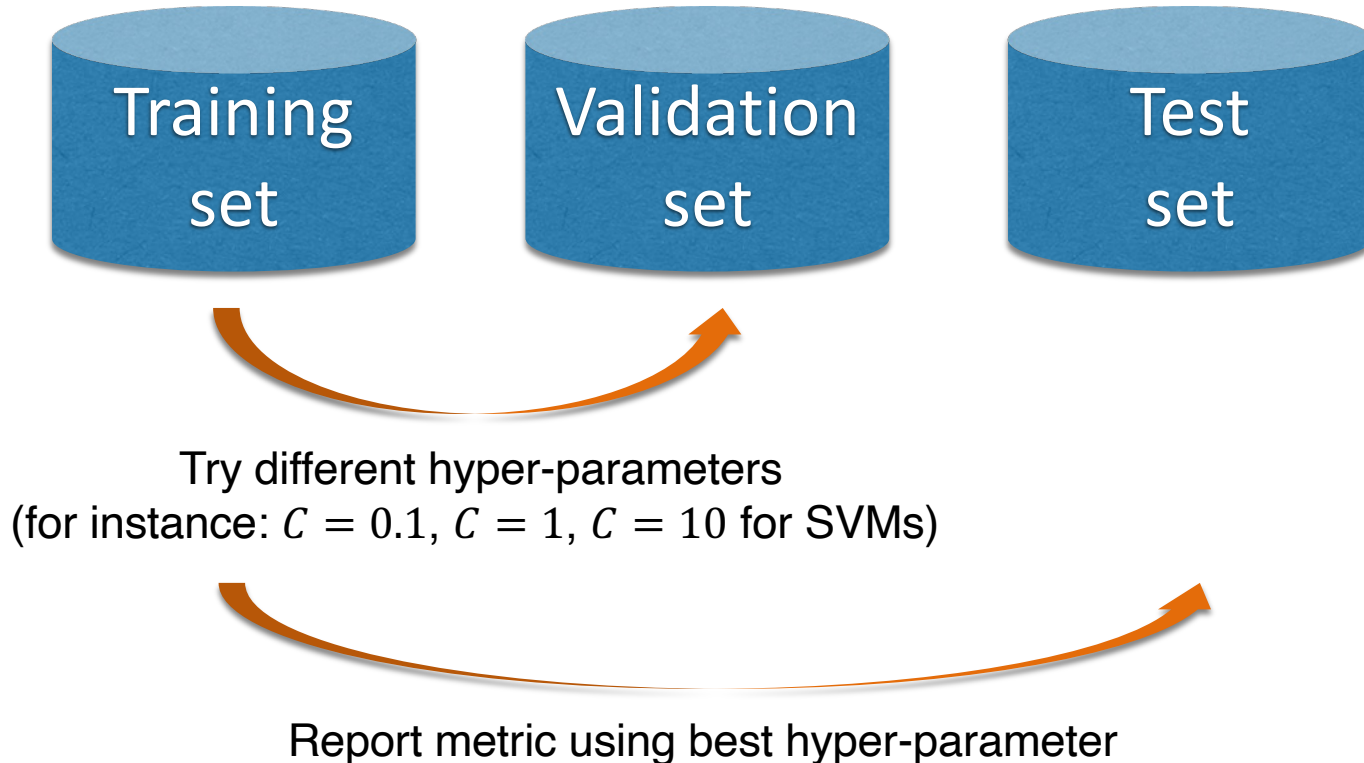
- Overfitting:
 - * More complex methods fit better the training data (e.g., linear kernel versus cubic kernel)
 - * Some hyper-parameter values might also fit better the training data
 - * Usually poor performance in unseen data



- To prevent overfitting, how to “see” unseen data?
 - * Simulate it !

Training, validation, testing

- Consider three disjoint parts



- We can and should repeat this several times
 - * (discussion on variability coming next)

k-Fold Cross Validation

- Split training data D into k disjoint sets $S_1, \dots, S_k$
 - * Either randomly, or in a fixed fashion
 - * If D has n samples, then each fold has approximately n/k samples
 - * Popular choices: $k = 5$, $k = 10$, $k = n$ (leave-one-out)
- For $i = 1 \dots k$:
 - * train with sets $S_1, \dots, S_{i-1}, S_{i+1}, \dots, S_k$
 - * test on set S_i
 - * let M_i be the test metric (e.g., accuracy, MSE)
- Mean $\hat{\mu} = \frac{1}{k} \sum_{i=1}^k M_i$ variance $\hat{\sigma}^2 = \frac{1}{k} \sum_{i=1}^k (M_i - \hat{\mu})^2$

0.632 Bootstrapping

- Let $B > 0$, and n be the number of training samples in D
- For $i = 1 \dots B$:
 - * Pick n samples from D with replacement, call it S_i (S_i might contain the same sample more than once)
 - * train with set S_i
 - * test on the remaining samples ($D - S_i$)
 - * let M_i be the test metric (e.g., accuracy, MSE)
- Mean $\hat{\mu} = \frac{1}{B} \sum_{i=1}^B M_i$ variance $\hat{\sigma}^2 = \frac{1}{B} \sum_{i=1}^B (M_i - \hat{\mu})^2$

0.632 Bootstrapping

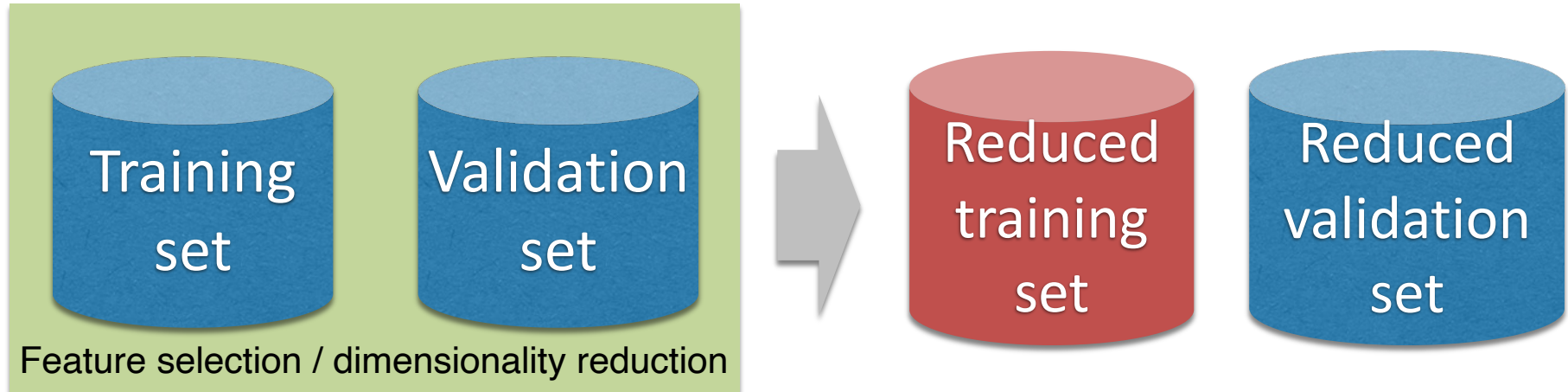
- Why 0.632?
- Recall that:
 - * We pick n items with replacement from out of n items
 - * We choose uniformly at random
- The probability of:
 - * not picking one particular item in 1 draw is $1 - 1/n$
 - * not picking one particular item in n draws is $(1 - 1/n)^n$
 - * picking one particular item in n draws is $1 - (1 - 1/n)^n$
- Finally: $\lim_{n \rightarrow \infty} 1 - (1 - 1/n)^n = 1 - 1/e \approx 0.632$

Nested cross-validation

- Useful to choose the best model, for instance, for hyperparameter tuning
- Inner cross-validation (e.g., 0.632 bootstrapping or training/validation) to try different hyperparameters
- Outer cross-validation (e.g., k-folds or training/test) to report metric using the best hyperparameter

Feature selection and cross-validation

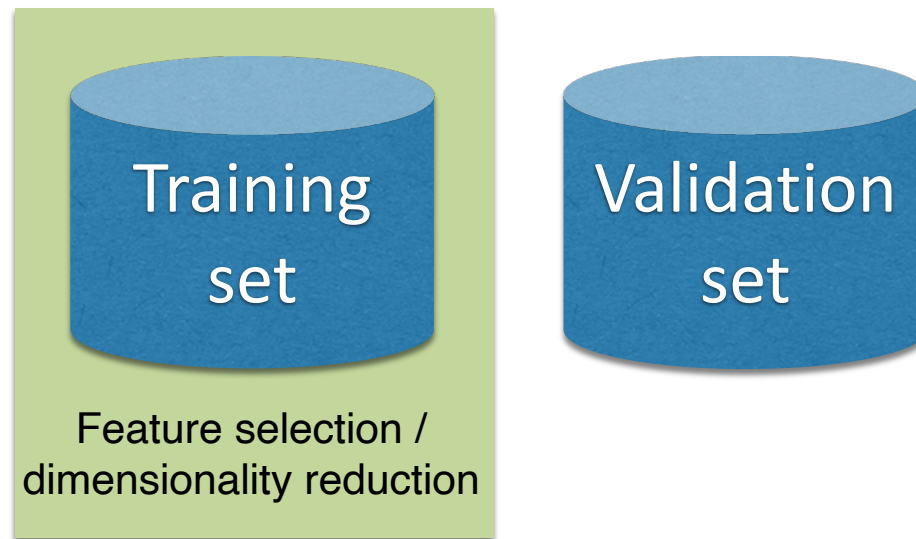
- **Incorrect way:** do not do feature selection (or dimensionality reduction) on the whole dataset, and then cross-validation



- Feature selection and dimensionality reduction on the whole dataset destroys cross-validation
 - * reduced training set would depend on the validation set
 - * thus, training is looking at the supposedly “unseen” data

Feature selection and cross-validation

- **Correct way:** feature selection (or dimensionality reduction) inside cross-validation, only applied to the training set



Mini Summary

- Different ways to perform cross-validation
 - * Train, validation, testing
 - * k-fold cross-validation
 - * 0.632 bootstrapping
 - * Nested cross-validation
 - * Be careful with feature selection
- Other considerations: keep similar characteristics (e.g., class proportion) on each data partition

Next: Variability

Variability

... or the need for several repetitions

Variability

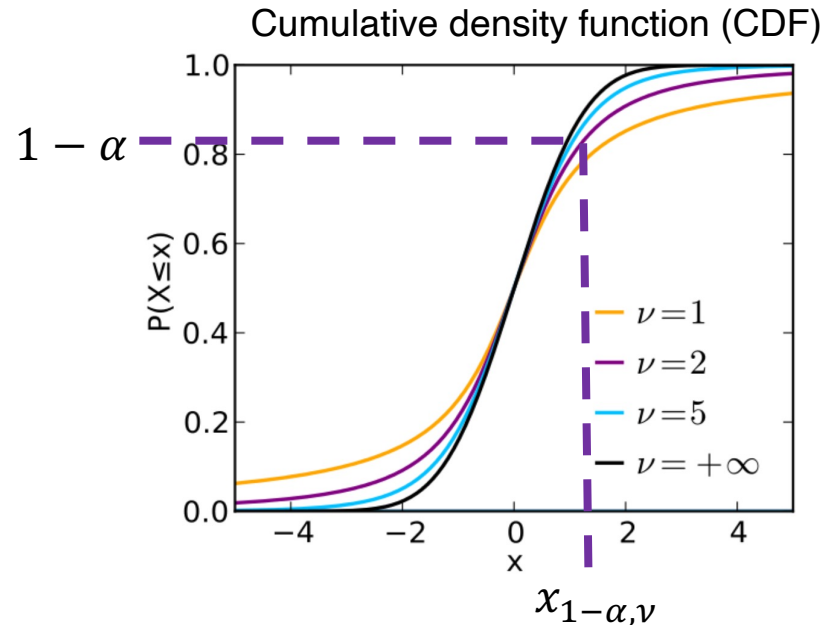
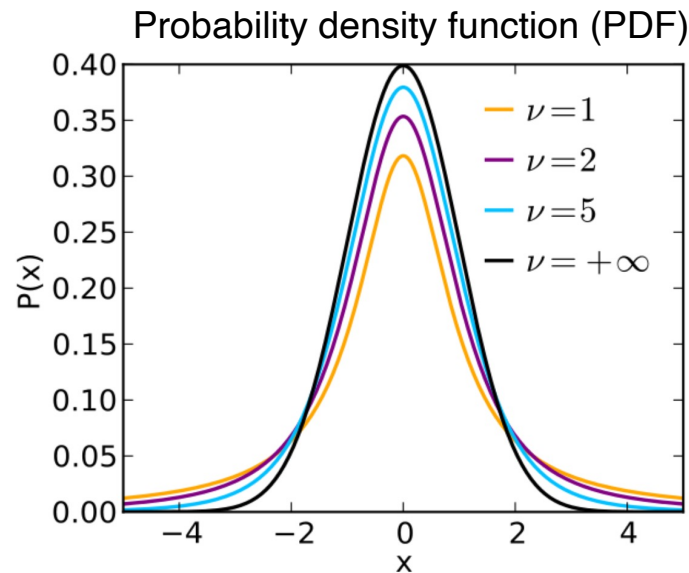
- When reporting any result, we cannot just report the mean of a test metric (e.g., k -fold cross validation, bootstrapping), we need to also report the variance
 - * Better way to compare alternatives
- Statistical hypothesis testing
 - * An imperfect technique applied on an imperfect setting, but very useful anyway to compare two alternatives
- Error bars
 - * Another way to report metric mean and variance

Statistical hypothesis testing

- How to compare two algorithms?
 - * Not only means, also variances!
 - * Here m is the number of repetitions, e.g., k for k -fold cross validation, or B for 0.632 bootstrapping
- Let $\hat{\mu}_1$ $\hat{\mu}_2$ $\hat{\sigma}_1^2$ $\hat{\sigma}_2^2$ be mean and variance of algorithms 1 and 2.
- When to reject null hypothesis $\mu_1 = \mu_2$ in favor of $\mu_1 > \mu_2$?
- Let $x = \frac{(\hat{\mu}_1 - \hat{\mu}_2)\sqrt{m}}{\sqrt{\hat{\sigma}_1^2 + \hat{\sigma}_2^2}}$ $v = \left\lceil \frac{(\hat{\sigma}_1^2 + \hat{\sigma}_2^2)^2 (m-1)}{\hat{\sigma}_1^4 + \hat{\sigma}_2^4} \right\rceil$
 Degrees of freedom of Student's t-distribution

Statistical hypothesis testing

- Student's t-distribution:



- For significance level α , degrees of freedom ν
 - * Find the value $x_{1-\alpha, \nu}$ for which $\text{CDF} = 1 - \alpha$
- If $x > x_{1-\alpha, \nu}$ reject null hypothesis $\mu_1 = \mu_2$ in favor of $\mu_1 > \mu_2$

Hypothesis testing: example 1

- Two algorithms tested with 9-fold cross validation
- Percentage of error on each left-out fold:
 - * A1: 11, 7, 13, 12, 12, 9, 10, 7, 10 $\hat{\mu}_1 = 10.1$ $\hat{\sigma}_1^2 = 4.1$
 - * A2: 10, 8, 12, 10, 11, 9, 13, 7, 9 $\hat{\mu}_2 = 9.9$ $\hat{\sigma}_2^2 = 3.2$
- Can we reject null hypothesis ($\mu_1 = \mu_2$) in favor of alternate hypothesis ($\mu_1 > \mu_2$) at 5% significance level?

$$x = \frac{(\hat{\mu}_1 - \hat{\mu}_2)\sqrt{m}}{\sqrt{\hat{\sigma}_1^2 + \hat{\sigma}_2^2}} = \frac{(10.1 - 9.9)\sqrt{9}}{\sqrt{4.1 + 3.2}} \approx 0.22$$

$$v = \left\lceil \frac{(\hat{\sigma}_1^2 + \hat{\sigma}_2^2)^2(m-1)}{\hat{\sigma}_1^4 + \hat{\sigma}_2^4} \right\rceil = \left\lceil \frac{(4.1 + 3.2)^2 8}{4.1^2 + 3.2^2} \right\rceil \approx \lceil 15.8 \rceil = 16$$

- Inverse CDF $x_{1-0.05,v} = x_{0.95,16} = 1.75$
- $x = 0.22 \leq 1.75 = x_{0.95,16}$ then cannot reject null

Hypothesis testing: example 2

- Two algorithms tested with 9-fold cross validation
- Percentage of error on each left-out fold:
 - * A1: 10, 12, 14, 13, 13, 10, 11, 10, 11 $\hat{\mu}_1 = 11.6$ $\hat{\sigma}_1^2 = 2$
 - * A2: 10, 8, 12, 10, 11, 9, 13, 7, 9 $\hat{\mu}_2 = 9.9$ $\hat{\sigma}_2^2 = 3.2$
- Can we reject null hypothesis ($\mu_1 = \mu_2$) in favor of alternate hypothesis ($\mu_1 > \mu_2$) at 5% significance level?

$$x = \frac{(\hat{\mu}_1 - \hat{\mu}_2)\sqrt{m}}{\sqrt{\hat{\sigma}_1^2 + \hat{\sigma}_2^2}} = \frac{(11.6 - 9.9)\sqrt{9}}{\sqrt{2 + 3.2}} \approx 2.24$$

$$v = \left\lceil \frac{(\hat{\sigma}_1^2 + \hat{\sigma}_2^2)^2(m-1)}{\hat{\sigma}_1^4 + \hat{\sigma}_2^4} \right\rceil = \left\lceil \frac{(2 + 3.2)^2 8}{2^2 + 3.2^2} \right\rceil \approx [15.2] = 16$$

- Inverse CDF $x_{1-0.05,v} = x_{0.95,16} = 1.75$
- $x = 2.24 > 1.75 = x_{0.95,16}$ then reject null

Error bars

- Two algorithms tested with 9-fold cross validation

- Percentage of error on each left-out fold:

* A1: 11, 7, 13, 12, 12, 9, 10, 7, 10

$$\hat{\mu}_1 = 10.1 \quad \hat{\sigma}_1^2 = 4.1$$

* A2: 10, 8, 12, 10, 11, 9, 13, 7, 9

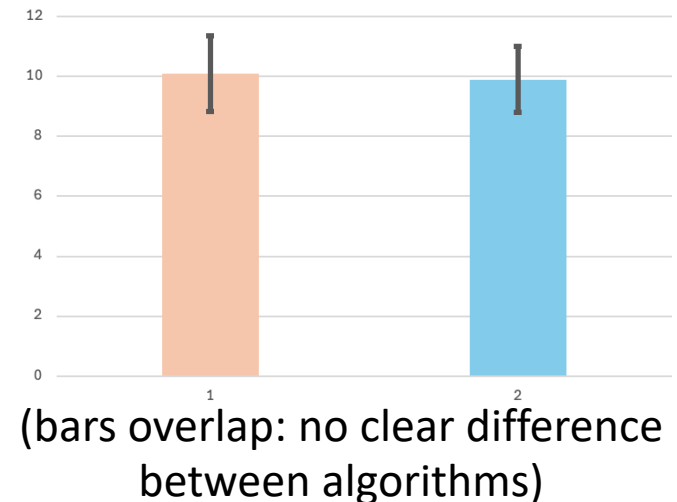
$$\hat{\mu}_2 = 9.9 \quad \hat{\sigma}_2^2 = 3.2$$

- Error bar $\hat{\mu}_j \pm \frac{\hat{\sigma}_j}{\sqrt{m}} x_{1-\alpha/2, m-1}$

* at 10% significance level

* $10.1 \pm \frac{\sqrt{4.1}}{\sqrt{9}} 1.86$ which is 10.1 ± 1.26

* $9.9 \pm \frac{\sqrt{3.2}}{\sqrt{9}} 1.86$ which is 9.9 ± 1.11



Mini Summary

- We need to report metrics' mean and variance
- Use statistical hypothesis testing and/or error bars to compare alternatives

Next: What is a sample?

What is a sample?

... a topic that need to be discussed
more often

What is a sample?

- In this lecture we assume that each sample is a different “unit of interest” for the experimenter
- Never sample the same “unit of interest” several times
 - * In a medical application, we might be interested on knowing the accuracy (and variance) with respect to patients.
 - * Taking two visits of the same patient as two different samples would be incorrect.
- Collect more data, if necessary
 - * Never duplicate data as a means to claim that you have more data samples, since your data will not capture the right variability, e.g., taking 1000 pictures of 3 objects versus taking one picture, each for one of 3000 objects.

Mini Summary

- Different performance metrics
- Different ways to perform cross-validation
- Reporting variability is important to compare alternatives

Next: Learning with experts